

Stephanie Wang

Brief intro

I have worked in and published papers in the fields of **geometric deep learning**, **geometry optimization**, and **physics-based simulation** of all kinds of materials (solids, fluids, waves, cloth, hair, friction / contact, deformable bodies, plasticity, visco-elasto-plasticity, and fracture / damage). My research aims to provide mathematical and engineering solutions to bring the virtual world to reality by building robust and high-performance computer graphics algorithms.

Education

Ph.D. and M.S. in Mathematics, *UCLA*, Eugene V. Cota-Robles Fellow. **2014-2020**
Committee: [Jeffrey D. Eldredge](#), [Wotao Yin](#), [Luminita Aura Vese](#), and [Joseph M. Teran](#) (advisor)
B.S. in Mathematics, *National Taiwan University*, *magna cum laude*. **2009-2013**

Experience

Research

Research Engineer – reporting to Dr. Qingnan Zhou, *Adobe*, Seattle, WA. **2024-present**
Geometry and physics research and software development, with a focus on geometry processing in machine learning and 3D modeling.

Postdoc – with Prof. Albert Chern, *UCSD*, San Diego, CA. **2020-2023**
Geometry processing and physical simulation using mathematical insights from geometric measure theory, exterior calculus, partial differential equations, and optimization theory. Mentored students: [Mohammad Sina Nabizadeh](#), [Shiyang Jia](#), [Chad McKell](#), [Hang Yin](#), [Baichuan Wu](#).

(Note: I took a full-time medical leave between Feb-Aug 2022 to recover from an acute illness.)

Ph.D. Study – with Prof. Wilfrid Gangbo, *UCLA*, Los Angeles, CA. **2019-2020**
Regularity theory for minimizers of polyconvex functionals related to Navier-Stokes equation.

Exchange Study – with Prof. Johan Gaume, *EPFL*, Lausanne, Switzerland. **2019 summer**
Simulations and data analysis of snow and tire interaction, avalanche release, and snow micro-structure.

Ph.D Study – with Prof. Joseph Teran, *UCLA*, Los Angeles, CA. **2016-2019**
Physics-based simulations of various materials with Material Point Method and Finite Element Method, using continuum mechanics, convex and nonconvex optimization technique, numerical analysis, parallel computing.

Tech Intern, *Walt Disney Animation Studio*, Burbank, CA. **2018 summer**
R&D for pioneering simulation technology in animated feature films, teaming with VFX artists.

Research Assistant – with Prof. Wen-Wei Lin, *NCTU*, Hsinchu, Taiwan. **2013-2014**
Generalized eigenvalue problems using MATLAB programming.

Teaching

Assistant Adjunct Professor, *UCLA Math Dept*, Los Angeles, CA (virtual). **2020**
Taught remote classes for upper and lower division undergraduate courses: Machine Learning (Math156) and Calculus of Several Variables (Math32A).

Graduate Student Instructor, *UCLA Math Dept*, Los Angeles, CA. **2019 spring**
Taught course: Linear Algebra and Applications (Math33A).

Teaching Assistant, *UCLA Math Dept*, Los Angeles, CA. **2015-2020**
Led discussion sessions and graded homework/exams for 11 undergraduate and graduate level courses: linear algebra and introduction to mathematical proofs (Math 115A), undergrad- and grad-level numerical methods (Math 151B, 269A), introductory, intermediate, and advanced C++ programming (PIC 10A, 10B, 10C).

Awards

Rising Stars in Computer Graphics Research, <i>WiGRAPH</i>.	May 2022
Best Paper Award, <i>SCA</i> 2019.	Jul 2019
Eugene V. Cota-Robles Fellowship, <i>UCLA</i>.	Sep 2014
Dean's Award, College of Science, <i>National Taiwan University</i>.	Jun 2013
Bronze Medal in Applied and Computational Mathematics, <i>S.T. Yau College Student Mathematics Contest</i>.	Aug 2012

Preprint

Wave Simulations in Infinite Spacetime

Chad McKell, Mohammad Sina Nabizadeh, [Stephanie Wang](#), Albert Chern

Publications

Implicit UVs: Real-time Semi-global Parameterization of Implicit Surfaces

Baptiste Genest, Pierre Gueth, J  r  my Levallois, [Stephanie Wang](#)

Computer Graphics Forum (Eurographics 2025)

Physical Cyclic Animations

Shiyang Jia, [Stephanie Wang](#), Tzu-Mao Li, Albert Chern

Proceedings of the ACM on Computer Graphics and Interactive Techniques (SCA 2023)

Exterior Calculus in Graphics: Course Notes for a SIGGRAPH 2023 Course

[Stephanie Wang](#), Mohammad Sina Nabizadeh, Albert Chern

SIGGRAPH '23: ACM SIGGRAPH 2023 Courses

Fluid Cohomology

Hang Yin, Mohammad Sina Nabizadeh, Baichuan Wu, [Stephanie Wang](#), Albert Chern

ACM Transactions on Graphics (SIGGRAPH 2023)

Covector Fluids

Mohammad Sina Nabizadeh, [Stephanie Wang](#), Ravi Ramamoorthi, Albert Chern

ACM Transactions on Graphics (SIGGRAPH 2022)

DeepCurrents: Learning Implicit Representations of Shapes with Boundaries

David Palmer, Dmitriy Smirnov, [Stephanie Wang](#), Albert Chern, Justin Solomon

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2022)

Role Detection in Bicycle-Sharing Networks Using Multilayer Stochastic Block Models

Jane Carlen, Jaume de Dios Pont, Cassidy Mentus, Shyr-Shea Chang, [Stephanie Wang](#), Mason A. Porter

Network Science, 2022

Computing minimal surfaces with differential forms

[Stephanie Wang](#) and Albert Chern

ACM Transactions on Graphics (SIGGRAPH 2021)

Computational micromechanics of porous brittle solids

Lars Blatny, Henning L  we, [Stephanie Wang](#), Johan Gaume

Computers and Geotechnics, 2021

A Material Point Method for Elastoplasticity with Ductile Fracture and Frictional Contact

[Stephanie Wang](#)

UCLA Doctoral Dissertation, 2020

A thermomechanical material point method for baking and cooking

Mengyuan Ding, Xuchen Han, [Stephanie Wang](#), Theodore F. Gast, Joseph M. Teran

ACM Transactions on Graphics (SIGGRAPH Asia 2019)

A Hybrid Material Point Method for Frictional Contact with Diverse Materials

Xuchen Han, Theodore F. Gast, Qi Guo, [Stephanie Wang](#), Chenfanf   Jiang, Joseph M. Teran

Proceedings of the ACM on Computer Graphics and Interactive Techniques (SCA 2019)

Simulation and Visualization of Ductile Fracture with the Material Point Method

[Stephanie Wang](#), Mengyuan Ding, Theodore F. Gast, Leyi Zhu, Steven Gagniere, Chenfanfu Jiang, Joseph M. Teran

Proceedings of the ACM on Computer Graphics and Interactive Techniques (**SCA 2019 Best Paper**)

Invited talks

Conferences / Workshops

ICIAM 2023 , Tokyo, Japan.	Aug 2023
SIGGRAPH 2023 , Los Angeles, CA.	Aug 2023
Geometry Workshop in Obergurgl 2021 , Obergurgl, Austria.	Sep 2021
SIGGRAPH 2021 , (virtual).	Aug 2021
SCA 2019 , Los Angeles, CA.	Aug 2019

Colloquia / Seminars

NCSU , Raleigh, NC (virtual).	Feb 2022
MIT , Cambridge, MA.	Nov 2021
Autodesk , (virtual).	Nov 2021
Online Seminar Geometric Analysis , (virtual).	Nov 2021
Toronto Geometry Colloquium , Toronto, ON (virtual).	Oct 2021
UCSD (CSE Vis-Comp) , San Diego, CA (virtual).	Apr 2021
UCSD (CCoM) , San Diego, CA (virtual).	Jan 2021
CMU , Pittsburgh, PA (virtual).	Dec 2020
GAMES Webinar , (virtual).	May 2020
College of the Holy Cross , Worcester, MA (virtual).	Nov 2019
Inria Grenoble-Rhône-Alpes , Grenoble, France.	Sep 2019
ETH Zürich , Zürich, Switzerland.	Aug 2019

Graduate Student Seminars

EPFL , Lausanne, Switzerland.	Aug 2019
UCLA , Los Angeles, CA.	Nov 2018

Services

- Program committee**, *ACM SIGGRAPH, Eurographics*. **2022-present**
ACM SIGGRAPH 2025 technical papers program committee,
SCA 2024 international program committee,
ACM SIGGRAPH 2023 poster jury, and
Eurographics 2023 short paper program committee.
- Tertiary reviewer**, *ACM SIGGRAPHs, Eurographics, ICLAM, IEEE TVCG*. **2021-present**
- Research project mentor**, *MIT Summer Geometry Initiative*. **2021**
Designed a research project and advised undergraduate fellows on minimal surfaces using both Lagrangian and Eulerian representations.
- Math Dept Representative**, *Graduate Student Association, UCLA*. **2017-2020**
Advocated for student rights in campus-level organizations and organized cross-department social events.
- Volunteer**, *AWiSE STEM Day, Explore Your Universe*. **2015-2020**
Presented interactive math booth in annual science fair designated for middle school girls and general public.
- Chief Organizer**, *Women in Math, UCLA*. **2016-2018**
Organized social and volunteering events, represented and advocated for women in math dept.
- Creator**, *Women in Math Mentorship Program, UCLA*. **2017**
Negotiated for fundings and created the program that hosts regular mixers for undergraduate and graduate fellows to increase connection, awareness, and mentorship.
- Fellow Mentor**, *California Teach, UCLA*. **2016-2018**
Mentored and advised Math and Statistics undergraduate students from underrepresented demographics.
- Vice President**, *Lambda Club, National Taiwan University*. **2012-2013**
Organized academic and social events and grew the community from 3 people to 30+ during my service.

Skills

- Programming**: C++ (Eigen, tbb), Python (PyTorch, SciPy), lua, MATLAB (CVX), $\LaTeX$, zsh
- Tools**: Houdini, Vim, git, gdb, valgrind
- Math**: Optimization, differential geometry, numerical and theoretical PDEs, scientific computing
- Languages**: English and Mandarin Chinese (bilingual)
- Hobbies**: Hiking, cooking, and people watching

Last updated: May 20, 2025.